BDA Final — Mock Paper 2 · Solutions

Full step-by-step worked solutions · 100 marks

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Topics covered in this PDF:

- $s^{r} = 0$
- $1 - s^{r} = 0$
- $(1 - s^{r})^{b} = 0$

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Prepared as a comprehensive professor-grade study tutor.

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> **Note**
>
> Each marking point is shown in **[brackets]** so you can self-grade.

Section A — Short Numerical (20 marks)

A1. PCY Hash Buckets — Solution (5 marks)

(a) Hashing each pair with *h(i,j) = (i · j) bmod 11*:

Pair	*i · j*	*h(i,j) = i· j bmod 11*	Count
{1,4}	4	**4**	5
{2,3}	6	**6**	4
{2,5}	10	**10**	6
{3,5}	15	**4**	3
{1,6}	6	**6**	4
{4,5}	20	**9**	7
{3,4}	12	**1**	5
{2,4}	8	**8**	2

Now sum counts per bucket: **[2 marks]**

Bucket	Pairs in bucket	Total count
B0	—	0
B1	{3,4}	5
B2	—	0
B3	—	0
B4	{1,4}, {3,5}	**5 + 3 = 8**
B5	—	0
B6	{2,3}, {1,6}	**4 + 4 = 8**
B7	—	0
B8	{2,4}	2
B9	{4,5}	**7**
B10	{2,5}	6

[1 mark for correct totals]

(b) Frequent buckets (count ≥ 7): **B4, B6, B9**. **[1 mark]**

(c) Pair {3,5} hashes to bucket B4 which is **frequent**. PCY does mark {3,5} as a **candidate** for Pass 2 — *however* the pair only needs to **survive** Pass 2 if its actual support reaches the support threshold. Being in a frequent bucket is **necessary but not sufficient**: a frequent bucket can be inflated by other pairs (here {1,4} alone contributed 5/8 of the count). **[1 mark]**

A2. LSH S-curve Probability — Solution (5 marks)

(a) Standard LSH banding probability:

$Pr[candidate] = 1 - big(1 - s^r big)^b$.

[1 mark]

(b) With $s = 0.6$, $r = 5$, $b = 20$:

- $s^r = 0.6^5 = 0.07776$ **[1 mark]**
- $1 - s^r = 0.92224$ **[½ mark]**
- $(1 - s^r)^b = 0.92224^{20}$.

Taking logs: $ln(0.92224) = -0.080927 \Rightarrow 20 \cdot (-0.080927) = -1.61855 \Rightarrow 0.92224^{20} = e^{-1.61855} \approx 0.19821$ **[1½ marks]**

- $Pr = 1 - 0.19821 = 0.802$ (≈ 80.2 %) **[1 mark]**

A3. Recommender Baseline Prediction — Solution (5 marks)

(a) ^ $r_{xi} = \mu + b_x + b_i$. **[1 mark]**

(b) ^ $r_{xi} = 3.5 + 0.4 + (-0.2) = 3.7$. **[2 marks]**

(c) New prediction with $b_j = +0.5$:

$$\hat{r}_{xj} = 3.5 + 0.4 + 0.5 = 4.4.$$

Change = $4.4 - 3.7 = +0.7$. **[2 marks]**

(Or: change in item bias alone = $b_j - b_i = 0.5 - (-0.2) = 0.7$.)

A4. Spider Trap vs Dead-End — Solution (5 marks)

- **G1:** A *leftrightarrow* B, $C \to A$. The set \$A, B\$ is a **spider trap** — once the random surfer enters A or B they bounce only between the two; node C has no in-link from \$A,B\$, so probability mass leaks **into** the trap and never out. **[2 marks]**
- **G2:** C has out-degree 0 → **dead-end**. PageRank mass leaks out of the system entirely (must be patched by teleport / dead-end removal). **[2 marks]**
- **G3:** $A \to B \to C \to A$ — strongly connected, every node has an out-edge. **Neither** spider trap nor dead-end. **[1 mark]**

Section B — Medium Numerical (30 marks)

B1. MinHash + LSH — Solution (10 marks)

(a) Hash table for $x = 1..4$ [2 marks]

x	$h_1 = (x+1) \bmod 5$	$h_2 = (3x+1) \bmod 5$	$h_3 = (2x+3) \bmod 5$	$h_4 = (4x+1) \bmod 5$
1	2	4	0	0
2	3	2	2	4
3	4	0	4	3
4	0	3	1	2

(b) MinHash signatures. Take the **min over rows where doc = 1**. **[3 marks]**

- D_1 has 1's in \r_1, r_2\:
  - h_1: $min(2, 3) = 2$
  - h_2: $min(4, 2) = 2$
  - h_3: $min(0, 2) = 0$
  - h_4: $min(0, 4) = 0$
- D_2 has 1's in \r_2, r_3, r_4\:
  - h_1: $min(3, 4, 0) = 0$
  - h_2: $min(2, 0, 3) = 0$
  - h_3: $min(2, 4, 1) = 1$
  - h_4: $min(4, 3, 2) = 2$
- D_3 has 1's in \r_1, r_3, r_4\:
  - h_1: $min(2, 4, 0) = 0$
  - h_2: $min(4, 0, 3) = 0$
  - h_3: $min(0, 4, 1) = 0$
  - h_4: $min(0, 3, 2) = 0$

Signature matrix *SIG*:

	D_1	D_2	D_3
h_1	2	0	0
h_2	2	0	0
h_3	0	1	0
h_4	0	2	0

(c) LSH banding ($b = 2$, $r = 2$). [5 marks]

- **Band 1** (rows h_1, h_2):
  - $D_1 \rightarrow (2, 2)$
  - $D_2 \rightarrow (0, 0)$
  - $D_3 \rightarrow (0, 0)$

  - D_2 and D_3 collide → candidate pair (D_2, D_3).
- **Band 2** (rows h_3, h_4):
  - $D_1 \to (0, 0)$
  - $D_2 \to (1, 2)$
  - $D_3 \to (0, 0)$
  - D_1 and D_3 collide → candidate pair (D_1, D_3).

Candidate pairs after LSH: \$(D_1, D_3), (D_2, D_3)$\.

> **Note**
>
> *Sanity check:* True Jaccard similarities are $J(D_1,D_2) = 1/4$, $J(D_1,D_3) = 1/4$, $J(D_2,D_3) = 2/4 = 0.5$. The high-Jaccard pair (D_2,D_3) is correctly flagged; (D_1, D_3) is a (tolerable) false-positive at this small signature length.

B2. Spam Farm — Solution (10 marks)

Constants.

- $\beta = 0.85 \Rightarrow \beta^2 = 0.7225 \Rightarrow 1 - \beta^2 = 0.2775$
- $dfrac\beta1+\beta = dfrac0.851.85 = 0.45946$

(a) With $M = 10^5$, $N = 10^9$, $x = 10^{-4}$: **[5 marks]**

Term 1: $dfracx1 - \beta^2 = dfrac10^{-4}0.2775 = 3.6036 \times 10^{-4}$ **[2 marks]**

Term 2: $dfrac\beta1+\beta \cdot dfracMN = 0.45946 \cdot dfrac10^5 10^9 = 0.45946 \cdot 10^{-4} = 4.5946 \times 10^{-5}$ **[2 marks]**

$$y = 3.6036 \times 10^{-4} + 4.5946 \times 10^{-5} \approx \mathbf{4.063 \times 10^{-4}}.$$

[1 mark]

(b) Set $y^{star} = 10^{-3}$ and solve for M: **[5 marks]**

$$10^{-3} = 3.6036 \times 10^{-4} + 0.45946 \cdot \frac{M}{10^9}.$$

$$0.45946 \cdot \frac{M}{10^9} = 10^{-3} - 3.6036 \times 10^{-4} = 6.3964 \times 10^{-4}.$$

[2 marks]

$$\frac{M}{10^9} = \frac{6.3964 \times 10^{-4}}{0.45946} = 1.3922 \times 10^{-3}.$$

[2 marks]

boxed $M \approx 1.39 \times 10^6$ *farm pages.*

[1 mark]

> **Note**
>
> Roughly **14× more farm pages** than the original 10^5 to multiply the target rank by ~2.5.

B3. BFR — Solution (10 marks)

(a) Centroid. [2 marks]

boldsymbol c = fracSUMN = left(frac205, frac155right) = (4, 3).

(b) Variances and standard deviations. [3 marks]

$$\sigma_x^2 = \frac{\text{SUMSQ}_x}{N} - \left(\frac{\text{SUM}_x}{N}\right)^2 = \frac{90}{5} - 4^2 = 18 - 16 = 2.$$

$$\sigma_y^2 = \frac{50}{5} - 3^2 = 10 - 9 = 1.$$

$$\sigma_x = \sqrt{2} \approx 1.414, \qquad \sigma_y = 1.$$

(c) Mahalanobis distance for $P = (6, 5)$. **[3 marks]**

d^2(P,boldsymbol c) = left(frac6 - 4√2right)^2 + left(frac5 - 31right)^2 = frac42 + 4 = 2 + 4 = 6.

d(P,boldsymbol c) = √6 ≈ 2.449.

(d) Decision. Threshold = 3. Since $d \approx 2.449 < 3$, point P **joins the DS** cluster. **[2 marks]**

> **Note**
>
> The DS summary then updates to $N=6$, $SUM=(26, 20)$, $SUMSQ=(126, 75)$.

Section C — Large Multi-Part Numerical (50 marks)

C1. Hierarchical Clustering + CURE — Solution (25 marks)

Recap of points:

	P_1	P_2	P_3	P_4	P_5	P_6	P_7
	(1,1)	(1.5,1.5)	(5,5)	(3,4)	(4,4)	(10,10)	(9,9.5)

(a) Initial 7 × 7 distance matrix (upper triangle) — 5 marks

Using $d(p,q) = \sqrt{(p_x-q_x)^2+(p_y-q_y)^2}$:

	P_1	P_2	P_3	P_4	P_5	P_6	P_7
P_1	0	**0.707**	5.657	3.606	4.243	12.728	11.673
P_2		0	4.950	2.915	3.536	12.021	10.966
P_3			0	2.236	**1.414**	7.071	6.021
P_4				0	**1.000**	9.220	8.139
P_5					0	8.485	7.433

	P_1	P_2	P_3	P_4	P_5	P_6	P_7
P_6						0	**1.118**
P_7							0

Sample working (for full marks show 2–3 of these explicitly):

- $d(P_1,P_2) = \sqrt{0.5^2+0.5^2} = \sqrt{0.5} \approx 0.707$
- $d(P_3,P_5) = \sqrt{1+1} = \sqrt{2} \approx 1.414$
- $d(P_4,P_5) = \sqrt{1+0} = 1.000$
- $d(P_6,P_7) = \sqrt{1+0.25} = \sqrt{1.25} \approx 1.118$

[5 marks: 1 per row of explicit working + 1 for full matrix]

(b) Single-linkage agglomerative — first 4 merges — 10 marks

> **Note**
>
> Single-link: $d(C_a, C_b) = \min_{p \in C_a,\, q \in C_b} d(p, q)$.

Merge 1. Minimum entry = $d(P_1, P_2) = 0.707$. → Form cluster $X = \{P_1, P_2\}$. **[2 marks]**

Remaining clusters: X, $\{P_3\}$, $\{P_4\}$, $\{P_5\}$, $\{P_6\}$, $\{P_7\}$.

Updated single-link distances from X to singletons (min of two old rows): $d(X, P_3) = \min(5.657, 4.950) = 4.950$, $d(X, P_4) = \min(3.606, 2.915) = 2.915$, $d(X, P_5) = \min(4.243, 3.536) = 3.536$, $d(X, P_6) = \min(12.728, 12.021) = 12.021$, $d(X, P_7) = \min(11.673, 10.966) = 10.966$.

Other distances are unchanged.

Merge 2. Smallest remaining entry = $d(P_4, P_5) = 1.000$. → Form cluster $A = \{P_4, P_5\}$. **[2 marks]**

Update distances from A to others: $d(A, X) = \min(2.915, 3.536) = 2.915$, $d(A, P_3) = \min(2.236, 1.414) = 1.414$, $d(A, P_6) = \min(9.220, 8.485) = 8.485$, $d(A, P_7) = \min(8.139, 7.433) = 7.433$.

Merge 3. Smallest remaining entry = $d(P_6, P_7) = 1.118$. → Form cluster $Z = \{P_6, P_7\}$. **[2 marks]**

Update distances from Z: $d(Z, X) = \min(12.021, 10.966) = 10.966$, $d(Z, A) = \min(8.485, 7.433) = 7.433$, $d(Z, P_3) = \min(7.071, 6.021) = 6.021$.

Merge 4. Current cluster set: X, A, $\{P_3\}$, Z.

Pairwise distances:

pair	distance
X–A	2.915
X–P_3	4.950
X–Z	10.966
A–P_3	**1.414**

pair	distance
A–Z	7.433
P_3–*Z*	6.021

Minimum = $d(A, P_3) = 1.414$. → Merge $\{P_3\}$ into *A*, giving cluster $B = \{P_3, P_4, P_5\}$. **[2 marks]**

Cluster set after merge 4 (3 clusters):

$$X = \{P_1, P_2\}, \quad B = \{P_3, P_4, P_5\}, \quad Z = \{P_6, P_7\}.$$

[2 marks]

(c) CURE Phase 1 — 7 marks

Cluster $X = \{P_1, P_2\}$.

- Centroid $\boldsymbol{c}_X = (1.25, 1.25)$.
- Distance from each point to $\boldsymbol{c}_X$: both equal $\sqrt{0.0625+0.0625} = \sqrt{0.125} \approx 0.354$. Tie → pick P_1 first.
- Second rep = farthest from $\{P_1\} = P_2$ (only other choice, distance 0.707).
- Reps before shrinking: $\{P_1, P_2\}$.
- Shrink with $\alpha = 0.4 \rightarrow r' = 0.6\, r + 0.4\, \boldsymbol{c}_X$:
  - $P_1' = 0.6\,(1,1) + 0.4\,(1.25,1.25) = (0.6+0.5, 0.6+0.5) = (1.1, 1.1)$.
  - $P_2' = 0.6\,(1.5,1.5) + 0.4\,(1.25,1.25) = (0.9+0.5, 0.9+0.5) = (1.4, 1.4)$.

[2 marks]

Cluster $B = \{P_3, P_4, P_5\}$.

- Centroid $\boldsymbol{c}_B = \text{big}(\text{tfrac}5+3+43, \text{tfrac}5+4+43\text{big}) = (4, 4.333)$.
- Distances to centroid:
  - $d(\boldsymbol{c}_B, P_3) = \sqrt{1 + (5-4.333)^2} = \sqrt{1 + 0.444} = \sqrt{1.444} \approx 1.202$
  - $d(\boldsymbol{c}_B, P_4) = \sqrt{1 + (4-4.333)^2} = \sqrt{1.111} \approx 1.054$
  - $d(\boldsymbol{c}_B, P_5) = \sqrt{0 + 0.111} \approx 0.333$
  - Farthest = P_3 → first rep.
- Second rep = farthest from $\{P_3\}$ among the rest:
  - $d(P_3, P_4) = \sqrt{4+1} = \sqrt{5} \approx 2.236$
  - $d(P_3, P_5) = \sqrt{1+1} = \sqrt{2} \approx 1.414$
  - Farthest = P_4 → second rep.
- Reps before shrinking: $\{P_3, P_4\}$.
- Shrink:
  - $P_3' = 0.6\,(5,5) + 0.4\,(4, 4.333) = (3 + 1.6, 3 + 1.733) = (4.6, 4.733)$.
  - $P_4' = 0.6\,(3,4) + 0.4\,(4, 4.333) = (1.8 + 1.6, 2.4 + 1.733) = (3.4, 4.133)$.

[3 marks]

Cluster $Z = \{P_6, P_7\}$.

- Centroid $\boldsymbol{c}_Z = (9.5, 9.75)$.
- Distance from each point to $\boldsymbol{c}_Z$ both equal $\sqrt{0.25+0.0625} = \sqrt{0.3125} \approx 0.559$. Tie → pick P_6 first, then P_7.

- Reps before shrinking: \P_6, P_7\.
- Shrink:
  - $P_6' = 0.6\ (10,10) + 0.4\ (9.5, 9.75) = (6 + 3.8, 6 + 3.9) = (9.8, 9.9)$.
  - $P_7' = 0.6\ (9, 9.5) + 0.4\ (9.5, 9.75) = (5.4 + 3.8, 5.7 + 3.9) = (9.2, 9.6)$.

[2 marks]

Summary of all 6 representatives:

Cluster	Centroid	Rep 1 (shrunken)	Rep 2 (shrunken)
X	*(1.25, 1.25)*	*(1.1, 1.1)*	*(1.4, 1.4)*
B	*(4, 4.333)*	*(4.6, 4.733)*	*(3.4, 4.133)*
Z	*(9.5, 9.75)*	*(9.8, 9.9)*	*(9.2, 9.6)*

(d) CURE Phase 2 — Assigning *Q = (4.5, 4.2)* — 3 marks

Distances from *Q* to every shrunken representative:

Rep	Coords	Squared diff *(Delta x^2 + Delta y^2)*	Distance
P_1'	*(1.1, 1.1)*	$(3.4)^2 + (3.1)^2 = 11.56 + 9.61 = 21.17$	4.601
P_2'	*(1.4, 1.4)*	$(3.1)^2 + (2.8)^2 = 9.61 + 7.84 = 17.45$	4.177
P_3'	*(4.6, 4.733)*	$(0.1)^2 + (0.533)^2 = 0.01 + 0.284 = 0.294$	**0.543**
P_4'	*(3.4, 4.133)*	$(1.1)^2 + (0.067)^2 = 1.21 + 0.0045 = 1.2145$	1.102
P_6'	*(9.8, 9.9)*	$(5.3)^2 + (5.7)^2 = 28.09 + 32.49 = 60.58$	7.784
P_7'	*(9.2, 9.6)*	$(4.7)^2 + (5.4)^2 = 22.09 + 29.16 = 51.25$	7.159

Nearest rep = P_3' at distance ≈ **0.543**, which belongs to cluster *B*.

→ Assign *Q* to **cluster *B* = \P_3, P_4, P_5**. **[3 marks]**

C2. Item-Item CF + RMSE — Solution (25 marks)

(a) Pearson item similarities — 8 marks

Item means (over **only the rated cells** of that column):

Item	Ratings	Mean
M_1	5, 4, 3, 5	4.25
M_2	3, 4, 2, 3	3.00
M_3	5, 4, 3, 4	4.00
M_4	4, 4, 3, 5	4.00

[2 marks for the means]

sim(M_1, M_3). Common users (both items rated): U_2, U_4, U_5.

Centered ratings (subtract item mean):

User	M_1 — 4.25	M_3 — 4.00
U_2	-0.25	+1.00
U_4	-1.25	-1.00
U_5	+0.75	0.00

Dot product: $(-0.25)(1) + (-1.25)(-1) + (0.75)(0) = -0.25 + 1.25 + 0 = 1.000$.

$\|M_1\| = \sqrt{0.0625 + 1.5625 + 0.5625} = \sqrt{2.1875} \approx 1.479$. $\|M_3\| = \sqrt{1 + 1 + 0} = \sqrt{2} \approx 1.414$.

$$\text{sim}(M_1, M_3) = \frac{1.000}{1.479 \times 1.414} = \frac{1.000}{2.092} \approx \mathbf{0.478}.$$

[3 marks]

sim(M_2, M_3). Common users: U_3, U_4, U_5.

Centered ratings:

User	M_2 — 3.00	M_3 — 4.00
U_3	+1.00	0.00
U_4	-1.00	-1.00
U_5	0.00	0.00

Dot product: $(1)(0) + (-1)(-1) + (0)(0) = 1.000$.

$\|M_2\| = \sqrt{1 + 1 + 0} = \sqrt{2} \approx 1.414$. $\|M_3\| = \sqrt{0 + 1 + 0} = 1$.

$$\text{sim}(M_2, M_3) = \frac{1.000}{1.414 \times 1} \approx \mathbf{0.707}.$$

[3 marks]

(b) Predict ^ $r(U_1, M_3)$ — 8 marks

Items U_1 has rated: M_1, M_2, M_4. Need similarities to M_3 for all three.

We already have $sim(M_1,M_3) = 0.478$, $sim(M_2,M_3) = 0.707$.

sim(M_4, M_3). Common users: U_2, U_3, U_5.

Centered ratings:

User	M_4 — 4.00	M_3 — 4.00
U_2	0.00	+1.00
U_3	-1.00	0.00
U_5	+1.00	0.00

Dot product $= 0 + 0 + 0 = 0$.

$$\text{sim}(M_4, M_3) = 0.$$

[2 marks]

Pick top-*k* neighbours. $k = 2$, sort by absolute similarity:

Item	$	\mathrm{sim}(\cdot, M_3)	$	Rank
M_2	0.707	1		
M_1	0.478	2		
M_4	0.000	3 (drop)		

Neighbour set N_k = \M_1, M_2\. **[1 mark]**

Apply weighted-average:

$$\hat{r}(U_1, M_3) = \frac{0.707 \cdot r(U_1, M_2) + 0.478 \cdot r(U_1, M_1)}{0.707 + 0.478}.$$

Plug in $r(U_1, M_1) = 5$, $r(U_1, M_2) = 3$:

$$\hat{r}(U_1, M_3) = \frac{0.707 \cdot 3 + 0.478 \cdot 5}{1.185} = \frac{2.121 + 2.390}{1.185} = \frac{4.511}{1.185} \approx \mathbf{3.806}.$$

[5 marks: 2 formula + 3 numeric]

> **Note**
>
> Sanity: U_1 tends to rate high (5 and 4 elsewhere); 3.81 is plausibly "moderate-positive".

(c) RMSE — 5 marks

#	Predicted	Actual	Error ^ *r* - *r*	Squared
1	4	5	*-1*	1
2	3	4	*-1*	1
3	5	4	*+1*	1
4	3	4	*-1*	1
5	5	5	*0*	0

Sum of squared errors = *1+1+1+1+0 = 4*. Mean squared error = *4 / 5 = 0.8*.

$$\text{RMSE} = \sqrt{0.8} \approx \mathbf{0.894}.$$

[5 marks: 2 errors + 1 sum + 1 mean + 1 sqrt]

(d) Why Item-Item CF often beats User-User CF — 4 marks

A solid 3–4 sentence answer should hit:

1. **Stability of profiles.** An item's "personality" (which users like it) changes much more slowly than a user's tastes. Pre-computed item-item similarities therefore stay valid for days/weeks, while user-user similarities drift fast. **[1 mark]**
2. **Cardinality.** In production, |*users*| *gg* |*items*| (e.g. Netflix: hundreds of millions of users vs ~10^4–10^5 titles). The |*I*| × |*I*| similarity matrix is feasible to store (millions of entries) where the |*U*| × |*U*| matrix is not (quadrillions). **[1 mark]**
3. **Offline pre-computation.** Item-item similarities can be computed **once** offline and reused for many users; serving a prediction is a small weighted sum over the rated items. User-user requires recomputation as users' rating vectors evolve. **[1 mark]**

4. **Practical accuracy.** Items typically have many more ratings than any single user, giving denser, less-noisy similarity estimates. Empirical evaluations (Sarwar et al. 2001, Linden et al. Amazon 2003) show item-item ≥ user-user in MAE/RMSE. **[1 mark]**

Mark Summary

Section	Question	Marks
A	A1	5
A	A2	5
A	A3	5
A	A4	5
A subtotal		**20**
B	B1	10
B	B2	10
B	B3	10
B subtotal		**30**
C	C1	25
C	C2	25
C subtotal		**50**
Grand Total		**100**

— End of Mock Paper 2 Solutions —